Backward Error Analysis and Randomized Algorithms for Total Least Squares

Theory, Methods, and Reflections

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LS, TLS, and null-space estimation: connections in both directions

Development	least squares (LS)	total least squares (TLS)
Formulation	$\min_{\Delta b, x} \|\Delta b\|_2$ s.t. $Ax = b + \Delta b$	$\min_{\Delta A, \Delta b, x} \|[\Delta A, \Delta b]\|_{\mathrm{F}}$, s.t. $(A + \Delta A)x = b + \Delta b$
Normal equation	$A^\top A x = A^\top b$	$\left(A^\top A - \sigma_{n+1}^2 I\right) x = A^\top b$
Direct method	QR: $A = QR$ $x = R^\dagger Q^\top b$	SVD: $C = [A, b] = U \Sigma V^\top$, $x = -\dfrac{v_{n+1}[1:n]^\top}{v_{n+1}[n+1]}$
Solvers and techniques	Sketch-and-solve; Iterative refinement; ...	Relatively underdeveloped especially for large-scale systems
Null-space estimation: given $A \in \mathbb{R}^{m \times n}$ $(m \gg n)$, consider $\min_{V^\top V = I_k} \|AV\|_{\mathrm{F}}$.		

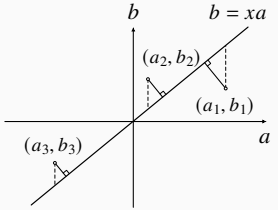


Fig: LS (dashed) vs. TLS (solid)

Research direction: Backward-stable LS solvers can inform TLS, but cannot be transferred directly: TLS has different structured normal equations and backward-error characterizations.

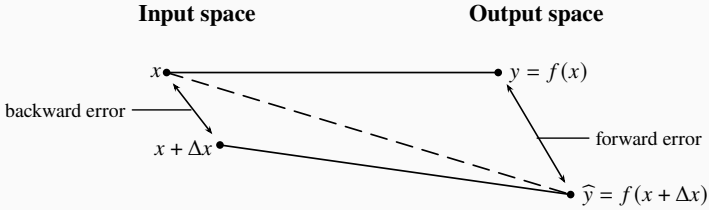
- Optimal TLS backward error and randomized algorithms [Shan and Wei 2025].

Research approach: Go beyond viewing TLS only as an application of null-space estimation. Use its structure and optimal backward error to address bottlenecks in eigensolvers such as LOBPCG.

- Efficient estimate for multidimensional TLS backward error [Shan and Wei 2026].

Backward error analysis: inform stability

Backward stability. A computed output $\widehat{y}$ is the exact answer to a nearby input $x + \Delta x$: $\widehat{y} = f(x + \Delta x)$, with an input perturbation at the scale of rounding error [Higham 2002].

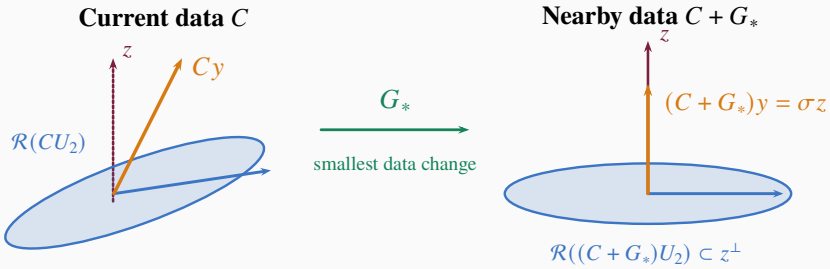


A small backward error certifies a nearby problem; a large forward error can still arise when that problem is ill-conditioned.

For TLS, let $\tau(\widetilde{x})$ be the smallest change in $C = [A, b]$ that makes $\widetilde{x}$ satisfy TLS stationarity. A normwise backward stable solver has $\tau(\widetilde{x})/\|C\|_{\mathrm{F}}$ at the scale of unit roundoff u.

Backward error analysis: act as a measure

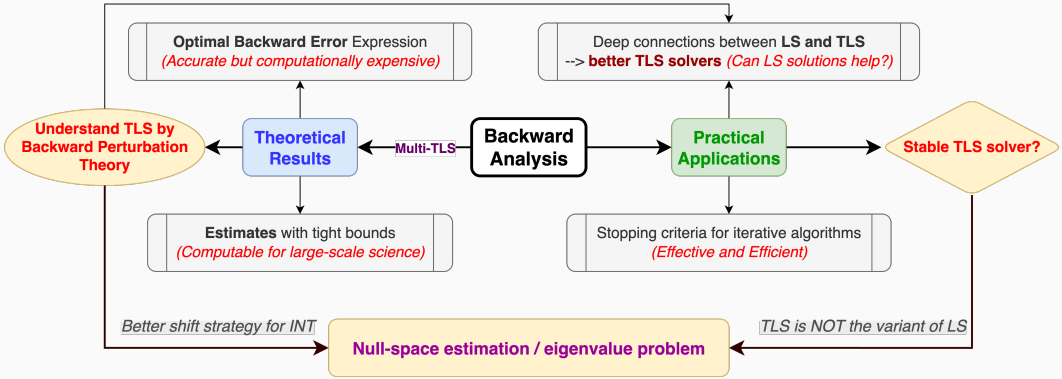
Consider the computed solution $\widetilde{x}$. Write $C = [A, b]$ and $y = [\widetilde{x}^\top, -1]^\top / \sqrt{1 + \|\widetilde{x}\|_2^2}$. Let U_2 span $y^\perp$ and $G_* = [E_*, f_*]$ be the optimal data perturbation [Shan 2026].



Geometry. $\mathcal{R}(CU_2)$ is the image of $y^\perp$ under C. For the perturbed matrix, $(C + G_*)y = \sigma z$ and $(C + G_*)y^\perp \subset z^\perp$; z is a unit left singular vector and σ its singular value.

Measure. When this is the unique smallest singular pair, $\tau(\widetilde{x}) = \|G_*\|_{\mathrm{F}}$ is the least data change that makes $\widetilde{x}$ an exact TLS solution. This perspective may inform singular-value and eigenvalue solvers.

Outline


Optimal Backward Error Expression
(Accurate but computationally expensive)
Deep connections between LS and TLS
--> better TLS solvers (Can LS solutions help?)
Understand TLS by Backward Perturbation Theory
Theoretical Results
Multi-TLS
Backward Analysis
Practical Applications
Stable TLS solver?
Estimates with tight bounds
(Computable for large-scale science)
Stopping criteria for iterative algorithms
(Effective and Efficient)
Better shift strategy for INT
Null-space estimation / eigenvalue problem
TLS is NOT the variant of LS

Backward perturbation for multidimensional TLS

mTLS problem. For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{m\times d}$, $m \gg n$,

$$\min_{\Delta A, \Delta B} \|[\Delta A, \Delta B]\|_{\mathrm{F}}, \qquad (A + \Delta A)X = B + \Delta B.$$

For a computed $\widetilde{X} \in \mathbb{R}^{n\times d}$, the θ-weighted backward error is

$$\eta_\theta(\widetilde{X}) = \min_{E,F} \big\{ \|[E, \theta F]\|_{\mathrm{F}} : \widetilde{X} \text{ solves the mTLS problem for data } (A + E, B + F) \big\}.$$

The nearby problem minimizes $\|[\Delta A, \Delta B]\|_{\mathrm{F}}$ subject to $(A + E + \Delta A)\widetilde{X} = B + F + \Delta B$.

1. **Theory:** derive an explicit expression for the mTLS backward error $\eta_\theta(\widetilde{X})$.
2. **Computation:** develop efficient estimates with tight bounds using randomization.

Backward perturbation for LS: related work

Single right-hand side. $\min_x \|Ax-b\|_2$, $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$. For a nonzero computed solution $\widehat{x}$ with $\widehat{r} = b - A\widehat{x}$, perturbations to A alone satisfy $\mathcal{E} = \mathcal{M}$ [Waldén et al. 1995]:

$$\mathcal{E} = \left\{ E \in \mathbb{R}^{m\times n} : \|b-(A+E)\widehat{x}\|_2 = \min_x \|b-(A+E)x\|_2 \right\},$$
$$\mathcal{M} = \left\{ \widehat{r}\widehat{x}^\dagger - vv^\dagger(A+\widehat{r}\widehat{x}^\dagger) + (I-vv^\dagger)Z(I-\widehat{x}\widehat{x}^\dagger) : v \in \mathbb{R}^m,\ Z \in \mathbb{R}^{m\times n} \right\}.$$

For changes to both A and b, define $\lambda = \lambda_{\min}(AA^\top - \widehat{r}\widehat{r}^\top/(1+\|\widehat{x}\|_2^2))$ and the feasible set

$$\mathcal{G} = \left\{ [E, f] : \|(b+f)-(A+E)\widehat{x}\|_2 = \min_x \|(b+f)-(A+E)x\|_2 \right\}.$$

The optimal error has the closed form [Karlson and Waldén 1997]:

$$\mu(\widehat{x}) = \min_{[E,f]\in\mathcal{G}} \|[E,f]\|_{\mathrm{F}} = \begin{cases} \|\widehat{r}\|_2/\|\widehat{x}\|_2, & \lambda \geq 0, \\ (\|\widehat{r}\|_2^2/\|\widehat{x}\|_2^2 + \lambda)^{1/2}, & \lambda < 0. \end{cases}$$

Direct evaluation is expensive and unstable. A stable alternative [Higham 2002] is

$$\mu(\widehat{x}) = \min\left\{ \frac{\|\widehat{r}\|_2}{\|\widehat{x}\|_2}, \sigma_{\min}[A, B] \right\}, \qquad B = \frac{\|\widehat{r}\|_2}{\|\widehat{x}\|_2}(I - \widehat{r}\widehat{r}^\dagger).$$

An efficient estimate is $\left\| (\|\widehat{x}\|_2^2 A^\top A + \|\widehat{r}\|_2^2 I)^{-1/2} A^\top \widehat{r} \right\|_2$ [Karlson and Waldén 1997; Gu 1998; Gratton et al. 2012; Grcar et al. 2008]. *Randomized LS solvers can now be fast, accurate, and stable* [Epperly et al. 2026].

Multiple right-hand sides: see [Hallman 2020].

Backward perturbation for TLS: related work

Single right-hand side (scaled TLS). For $\gamma > 0$,

$$\min_{\Delta A, \Delta b} \|[\Delta A, \gamma \Delta b]\|_{\mathrm{F}}, \qquad (A + \Delta A)x = b + \Delta b,$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $m \gg n$. For a computed $\widetilde{x} \in \mathbb{R}^n$, the θ-weighted backward error minimizes $\|[E, \theta f]\|_{\mathrm{F}}$ over nearby scaled TLS problems for which $\widetilde{x}$ is an exact solution.

Set $\widetilde{r} = b - A\widetilde{x}$ and

$$M = A(I - \widetilde{x}\widetilde{x}^{\dagger})A^{\top} - \frac{\theta^2 \widetilde{r}\widetilde{r}^{\top}}{1 + \theta^2 \|\widetilde{x}\|_2^2} + \frac{\theta^2 (A\widetilde{x} + \gamma^2 \|\widetilde{x}\|_2^2 b)(A\widetilde{x} + \gamma^2 \|\widetilde{x}\|_2^2 b)^{\top}}{\theta^2 \|\widetilde{x}\|_2^2 + \gamma^4 \|\widetilde{x}\|_2^4}.$$

The explicit formula in [Chang and Titley-Peloquin 2009] is

$$\eta_\theta^\gamma = \min_{E,f} \|[E, \theta f]\|_{\mathrm{F}}^2 = \frac{\theta^2 \|\widetilde{r}\|_2^2}{1 + \theta^2 \|\widetilde{x}\|_2^2} + \min\{0, \lambda_{\min}(M)\}.$$

The formula is explicit, but difficult to evaluate and interpret directly.

Our contribution. We extend the analysis to mTLS and connect its structure to LS.

A structural theorem: necessary and sufficient conditions for mTLS

Theorem [Shan and Wei 2026]. Let $C = [A, B]$, $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{m\times d}$, and $X \in \mathbb{R}^{n\times d}$. Assume $\sigma_n(C) > \sigma_{n+1}(C)$ and that V_{22} is nonsingular in $C = U\Sigma V^\top$. Then X is the mTLS solution if and only if

$$(A + BX^\top)^\top (AX - B) = 0$$

and, writing $\mathcal{R}(\cdot)$ for the range,

$$\max_{0\neq y\in\mathcal{R}\left(\left[\begin{smallmatrix} X \\ -I_d \end{smallmatrix}\right]\right)} \frac{\|Cy\|_2}{\|y\|_2} < \min_{0\neq y\in\mathcal{R}\left(\left[\begin{smallmatrix} I_n \\ X^\top \end{smallmatrix}\right]\right)} \frac{\|Cy\|_2}{\|y\|_2}.$$

Proof (key steps). Let $[T_1, T_2]$ be orthogonal:

$$T_1 = \begin{bmatrix} I_n \\ X^\top \end{bmatrix} (I_n + XX^\top)^{-1/2},$$
$$T_2 = \begin{bmatrix} X \\ -I_d \end{bmatrix} (I_d + X^\top X)^{-1/2}.$$

Then $\mathcal{A} = CT_1 = (A + BX^\top)(I_n + XX^\top)^{-1/2}$ and $R = CT_2 = (AX - B)(I_d + X^\top X)^{-1/2}$. The two conditions become $\mathcal{A}^\top R = 0$ and $\sigma_{\max}(R) < \sigma_{\min}(\mathcal{A})$.

$\mathcal{A}, R$ are indeed the essential components for TLS systems rather than the original A, b.

Remark: weighted TLS $\to$ LS. For one right-hand side, weight perturbations to b by $\gamma > 0$:

$$\eta^\gamma(x) = \frac{\gamma\|Ax - b\|_2}{\sqrt{1 + \gamma^2\|x\|_2^2}},$$

with structural normal equation

$$(A + \gamma^2 bx^\top)^\top (Ax - b) = 0.$$

$\gamma = 1$ gives TLS. As $\gamma \to 0$, $\eta_\gamma(x)/\gamma \to \|Ax - b\|_2$ and the equation becomes $A^\top(Ax - b) = 0$ (LS).

When only perturb the right-hand side

For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{m\times d}$, let $\widetilde{X} \in \mathbb{R}^{n\times d}$ have full column rank and $\widehat{R} = B - A\widetilde{X}$. The relaxed mTLS feasibility condition is

$$\mathcal{E}^{+}_{\mathrm{mTLS}} = \big\{E : ((A+E) + B\widetilde{X}^{\top})^{\top}[B - (A+E)\widetilde{X}] = 0\big\}.$$

Theorem. This set equals

$$\mathcal{M}^{+}_{\mathrm{mTLS}} = \big\{-YY^{\dagger}(A + B\widetilde{X}^{\top}) + (I - YY^{\dagger})\big[\widehat{R}\widetilde{X}^{\dagger} + Z(I - \widetilde{X}\widetilde{X}^{\dagger})\big] : Y \in \mathbb{R}^{m\times d}, \quad Z \in \mathbb{R}^{m\times n}\big\}.$$

Proof sketch: Define $\bar{A} = A + B\widetilde{X}^{\top}$ and $\bar{B} = B + B\widetilde{X}^{\top}\widetilde{X}$. Then

$$\underbrace{((A+E) + B\widetilde{X}^{\top})^{\top}[B - (A+E)\widetilde{X}]}_{\text{mTLS stationarity}} = (\bar{A}+E)^{\top}[\bar{B} - (\bar{A}+E)\widetilde{X}].$$

The right-hand side is the normal equation of the modified LS pair $(\bar{A}, \bar{B})$, with residual $\bar{B} - \bar{A}\widetilde{X} = \widehat{R}$. Applying the multiple-right-hand-side LS characterization [Sun 1996] gives the displayed parametrization of E.

Explicit optimal backward error for mTLS

Let $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{m\times d}$ and full-column-rank $\widetilde{X} \in \mathbb{R}^{n\times d}$. Define

$$\widetilde{A} = (A + B\widetilde{X}^\top)(I + \theta^{-2}\widetilde{X}\widetilde{X}^\top)^{-1/2},$$
$$\widetilde{R} = \theta(B - A\widetilde{X})(I + \theta^2\widetilde{X}^\top\widetilde{X})^{-1/2}.$$

For the eigenvalues $\lambda_1^{(\theta)} \le \cdots \le \lambda_m^{(\theta)}$ of $\widetilde{A}\widetilde{A}^\top - \widetilde{R}\widetilde{R}^\top$, the optimal error is

$$\eta_\theta(\widetilde{X}) = \left(\|\widetilde{R}\|_{\mathrm{F}}^2 + \lambda_*^{(\theta)}\right)^{1/2},$$

where

$$\lambda_*^{(\theta)} = \begin{cases} 0, & 0 \le \lambda_1^{(\theta)}, \\ \sum_{i=1}^{s} \lambda_i^{(\theta)}, & \lambda_s^{(\theta)} < 0 \le \lambda_{s+1}^{(\theta)},\ s < d, \\ \sum_{i=1}^{d} \lambda_i^{(\theta)}, & \lambda_d^{(\theta)} < 0. \end{cases}$$

Interpretation. The residual term $\|\widetilde{R}\|_{\mathrm{F}}^2$ is reduced by up to d negative eigenvalues. In particular, $\lambda_1^{(\theta)} \ge 0$ gives $\eta_\theta(\widetilde{X}) = \|\widetilde{R}\|_{\mathrm{F}}$. The rank-deficient and restricted-perturbation cases are treated in the paper [Shan and Wei 2026].

Practical computation: one right-hand side

Corollary: optimal backward error for TLS. With $d = 1$,

$$\tau_\theta(\widetilde{x}) = \min_{E,f} \|[E, \theta f]\|_{\mathrm{F}} = \left[\|\widetilde{r}\|_2^2 + \min\{0, \lambda_{\min}(\widetilde{A}\widetilde{A}^\top - \widetilde{r}\widetilde{r}^\top)\}\right]^{1/2},$$

where

$$\widetilde{A} = (A + b\widetilde{x}^\top)(I + \theta^{-2}\widetilde{x}\widetilde{x}^\top)^{-1/2}, \qquad \widetilde{r} = \theta(1 + \theta^2\|\widetilde{x}\|_2^2)^{-1/2}(b - A\widetilde{x}).$$

For numerical stability, this can be computed as

$$\tau_\theta(\widetilde{x}) = \min\{\|\widetilde{r}\|_2, \sigma_{\min}(M)\}, \qquad M = \begin{pmatrix} \widetilde{A}^\top \\ \|\widetilde{r}\|_2(I - \widetilde{r}\widetilde{r}^\dagger) \end{pmatrix}.$$

Theorem: computable estimate for TLS.

$$\widetilde{\tau}_\theta(\widetilde{x}) = \left\| (\widetilde{A}^\top \widetilde{A} + \|\widetilde{r}\|_2^2 I)^{-1/2} \widetilde{A}^\top \widetilde{r} \right\|_2, \qquad \widetilde{\tau}_\theta(\widetilde{x}) \le \tau_\theta(\widetilde{x}) \le \sqrt{2}\,\widetilde{\tau}_\theta(\widetilde{x}).$$

The SVD-based expression avoids subtracting close eigenvalues; the cheaper estimate stays within a factor $\sqrt{2}$ of the optimal error.

Sketching: a nearly isometric embedding

Definition: sketching matrix. A random $S \in \mathbb{R}^{s \times m}$ is a sketching matrix for $A \in \mathbb{R}^{m \times n}$ with distortion $\epsilon \in (0, 1)$ if

$$\forall y \in \mathbb{R}^n, \qquad (1-\epsilon)\|Ay\|_2 \le \|S(Ay)\|_2 \le (1+\epsilon)\|Ay\|_2.$$

For a random construction, this property holds with high probability.

Lemma: randomized preconditioning. Let A have full column rank and factor $SA = QK$, where $Q^\top Q = I$ and K is invertible. Then

$$\frac{1}{1+\epsilon} \le \sigma_{\min}(AK^{-1}) \le \sigma_{\max}(AK^{-1}) \le \frac{1}{1-\epsilon}.$$

Reason. Substitute $y = K^{-1}z$ in the embedding inequality and use $SAK^{-1} = Q$; the two bounds follow for every z. See [Woolfe et al. 2008].

Sketched TLS estimate: theorem and proof

Theorem [Shan and Wei 2025]. Suppose $S \in \mathbb{R}^{s\times m}$ embeds $\operatorname{span}([A, b])$ with distortion ϵ. For $z = \widetilde{A}^\top \widetilde{r}$ and $\rho = \|\widetilde{r}\|_2$, write

$$H = \widetilde{A}^\top \widetilde{A} + \rho^2 I, \quad H_S = (S\widetilde{A})^\top (S\widetilde{A}) + \rho^2 I, \quad \widetilde{\tau}_\theta^{\mathrm{sk}} = \sqrt{z^\top H_S^{-1} z}.$$

Then

$$(1-\epsilon)\widetilde{\tau}_\theta^{\mathrm{sk}} \le \tau_\theta(\widetilde{x}) \le \sqrt{2}(1+\epsilon)\widetilde{\tau}_\theta^{\mathrm{sk}}.$$

Proof (Rayleigh-quotient argument). Since $\operatorname{col}(\widetilde{A}) \subseteq \operatorname{span}([A, b])$, every nonzero v satisfies

$$\frac{v^\top H_S v}{v^\top H v} = \frac{\|S\widetilde{A}v\|_2^2 + \rho^2\|v\|_2^2}{\|\widetilde{A}v\|_2^2 + \rho^2\|v\|_2^2} \in [(1-\epsilon)^2, (1+\epsilon)^2].$$

Thus $(1-\epsilon)^2 H \preceq H_S \preceq (1+\epsilon)^2 H$. Inverting the inequalities and evaluating at z gives

$$(1-\epsilon)\widetilde{\tau}_\theta^{\mathrm{sk}} \le \widetilde{\tau}_\theta \le (1+\epsilon)\widetilde{\tau}_\theta^{\mathrm{sk}}.$$

Combine this with $\widetilde{\tau}_\theta \le \tau_\theta \le \sqrt{2}\,\widetilde{\tau}_\theta$ from the preceding slide. □

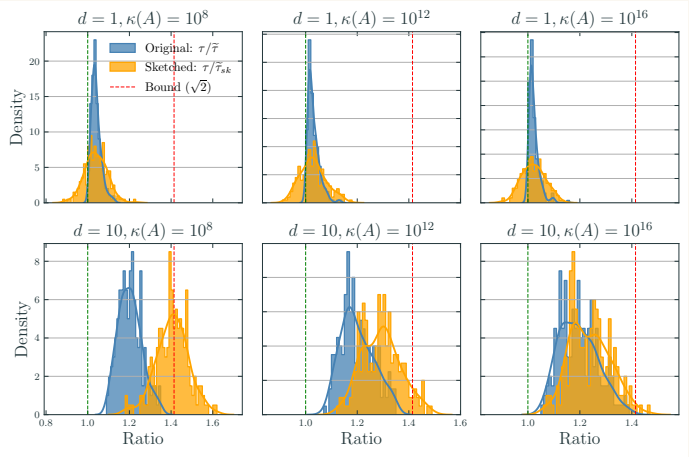
Algorithm

Input: $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, computed $\widetilde{x} \in \mathbb{R}^n$, and $\theta > 0$.
Output: $\widetilde{\tau}_\theta^{\mathrm{sk}}(\widetilde{x})$.

1. Draw an embedding $S \in \mathbb{R}^{s\times m}$ for $[A, b]$.
2. Set $\overline{A} \leftarrow A + b\widetilde{x}^\top$, $\widetilde{r} \leftarrow \theta(1+\theta^2\|\widetilde{x}\|_2^2)^{-1/2}(b - A\widetilde{x})$, and $\zeta^2 \leftarrow \widetilde{r}^\top\widetilde{r}$.
3. Apply the rank-one square-root update
$$\widetilde{A} \leftarrow \overline{A} + \left(\overline{A}\left(\frac{\theta}{\sqrt{\theta^2+\|\widetilde{x}\|_2^2}} - 1\right)\widetilde{x}\right)\frac{\widetilde{x}^\top}{\|\widetilde{x}\|_2^2}.$$
4. Compute $S\widetilde{A} = U_{\mathrm{sk}}\Sigma_{\mathrm{sk}}V_{\mathrm{sk}}^\top$, with singular values σ_i.
5. Evaluate the diagonal scaling:
$$\widetilde{\tau}_\theta^{\mathrm{sk}}(\widetilde{x}) \leftarrow \left\| \left[\frac{1}{\sqrt{\sigma_i^2+\zeta^2}}\right]_{i=1}^{n} \odot \left(V_{\mathrm{sk}}^\top \widetilde{A}^\top \widetilde{r}\right)\right\|_2.$$

Sparse sign choice. $S = \xi^{-1/2}[s_1, \ldots, s_m]$: each s_i has ξ randomly placed nonzeros with independent signs ± 1. The experiments use $s = 12n$ and $\xi = 8$.

Numerical test: tightness of the estimates



Construction ($\theta = 1$). $(m, n) = (3000, 20)$, $\kappa(A) \in \{10^8, 10^{12}, 10^{16}\}$, and $d \in \{1, 10\}$. Generate 200 instances per condition number, with logarithmically spaced singular values and Haar-random singular vectors.

Observation.
The direct estimate stays within the factor $\sqrt{2}$. Sketching remains accurate even at $\kappa(A) = 10^{16}$.

Practical computation: multiple right-hand sides

Theorem [Shan and Wei 2026]. Use $\widetilde{A}, \widetilde{R}$ from the optimal-error formula, and let $\widetilde{R} = W\Lambda Z^\top$ with singular values λ_j and left singular vectors w_j. Then

$$\nu_\theta(\widetilde{X})^2 = \sum_{j=1}^{r_{\widetilde{R}}} \lambda_j^2 \left\| (\widetilde{A}^\top \widetilde{A} + \lambda_j^2 I)^{-1/2} \widetilde{A}^\top w_j \right\|_2^2, \qquad \nu_\theta \le \eta_\theta \le \sqrt{2}\,\nu_\theta.$$

Proof, part 1: remove the orthogonality constraint. The optimal error has the equivalent form

$$\eta_\theta = \min\bigl\{ \|[M, N]\|_{\mathrm{F}} : \widetilde{A}^\top N + M^\top \widetilde{R} = -\widetilde{A}^\top \widetilde{R},\ M^\top N = 0 \bigr\}.$$

Dropping $M^\top N = 0$ gives the lower bound ν_θ. With $\widetilde{A} = U\Sigma V^\top$ and $c_{ij} = u_i^\top w_j$, the relaxed minimization separates into

$$\min_{m_{ij}, n_{ij}} (m_{ij}^2 + n_{ij}^2) \quad \text{s.t.} \quad \lambda_j m_{ij} + \sigma_i n_{ij} = -\sigma_i \lambda_j c_{ij}.$$

The minimum-norm solution is

$$m_{ij} = -\frac{\sigma_i^2 \lambda_j c_{ij}}{\sigma_i^2 + \lambda_j^2}, \qquad n_{ij} = -\frac{\sigma_i \lambda_j^2 c_{ij}}{\sigma_i^2 + \lambda_j^2}.$$

Its squared norm is $\sigma_i^2 \lambda_j^2 c_{ij}^2 / (\sigma_i^2 + \lambda_j^2)$; sum over i, j to obtain the stated formula.

Proof of the bounds (Continue)

From ν_θ to η_θ. Let (M, N) minimize the relaxed problem on the preceding slide. If $M^\top N = UDV^\top$, take $P_M = D^{1/2}U^\top$ and $P_N = -D^{1/2}V^\top$. Then $P_M^\top P_N = -M^\top N$ and $\|[P_M, P_N]\|_{\mathrm{F}}^2 = 2\|M^\top N\|_*$. Augment $\widetilde{A}, \widetilde{R}$ by zero rows and their perturbations by P_M, P_N; zero-row augmentation leaves η_θ unchanged. The new pair is feasible because

$$\left(\begin{bmatrix}\widetilde{A}\\ 0\end{bmatrix} + \begin{bmatrix}M\\ P_M\end{bmatrix}\right)^\top \left(\begin{bmatrix}\widetilde{R}\\ 0\end{bmatrix} + \begin{bmatrix}N\\ P_N\end{bmatrix}\right) = M^\top N + P_M^\top P_N = 0.$$

Consequently,

$$\eta_\theta^2 \le \nu_\theta^2 + 2\|M^\top N\|_* \le \nu_\theta^2 + 2\|M\|_{\mathrm{F}}\|N\|_{\mathrm{F}} \le 2\nu_\theta^2.$$

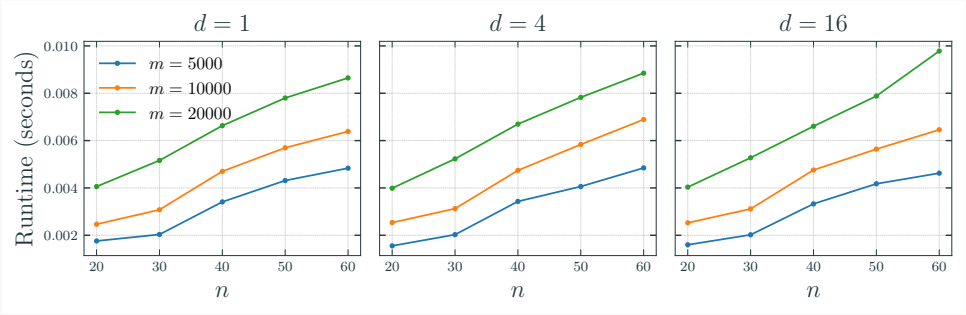
Sketching. If S embeds $\mathrm{span}([A, B])$ with distortion ϵ, replace $\widetilde{A}^\top \widetilde{A}$ by $(S\widetilde{A})^\top (S\widetilde{A})$ in ν_θ. For each j, let $H_j = \widetilde{A}^\top \widetilde{A} + \lambda_j^2 I$ and $H_{j,S} = (S\widetilde{A})^\top (S\widetilde{A}) + \lambda_j^2 I$. For $z_j = \widetilde{A}^\top w_j$, the sketched estimate is $\nu_\theta^{\mathrm{sk}\,2} = \sum_j \lambda_j^2 z_j^\top H_{j,S}^{-1} z_j$. The embedding argument gives

$$(1-\epsilon)^2 H_j \preceq H_{j,S} \preceq (1+\epsilon)^2 H_j \quad \Longrightarrow \quad (1-\epsilon)\nu_\theta^{\mathrm{sk}} \le \nu_\theta \le (1+\epsilon)\nu_\theta^{\mathrm{sk}}.$$

Therefore $1 - \epsilon \le \eta_\theta / \nu_\theta^{\mathrm{sk}} \le \sqrt{2}(1+\epsilon)$.

Numerical test: cost of the sketched mTLS estimate

Construction. Use the same random-matrix generator with $\kappa(A) = 10^6$, $m \in \{5, 10, 20\} \times 10^3$, $n \in \{20, 30, 40, 50, 60\}$, and $d \in \{1, 4, 16\}$. Measure the runtime of the sketched estimate as each dimension varies.

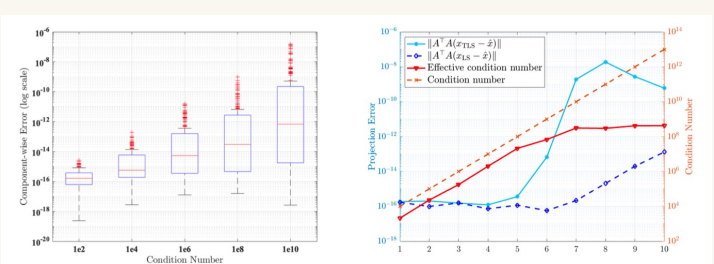


Observation. At fixed m and d, runtime grows approximately linearly with n; larger m raises the cost. The dependence on d is mild over the tested range.

Backward stability controls a modified TLS error

Let $A = U\Sigma V^\top$ with right singular vectors v_i and singular values σ_i. Write $\sigma_* = \sigma_{n+1}([A, b])$, $r_{\widehat{x}} = b - A\widehat{x}$, and let u denote unit roundoff. The componentwise bound in [Shan and Wei 2025] is

$$\left| v_i^\top \left[\left(1 - \frac{\sigma_*^2}{\sigma_i^2} \right) x_{\mathrm{TLS}} - \widehat{x} \right] \right| \lesssim \frac{\|\widehat{x}\|_2 \|[A, b]\|_2}{\sigma_i} u + \frac{\|r_{\widehat{x}}\|_2 \|[A, b]\|_2}{\sigma_i^2} u.$$



(a) Component-wise error

(b) Projection error

Backward stability controls this modified error, but closeness to x_{TLS} also needs σ_*/σ_i away from one. Here $\lesssim$ hides a low-degree polynomial factor in m, n.

LS residual, spectral gap, and TLS proximity

Assume A has full column rank and $x_{\mathrm{LS}} \neq 0$. Write $r_{\mathrm{LS}} = b - Ax_{\mathrm{LS}}$, $\sigma_* = \sigma_{n+1}([A, b])$, and $\delta = \sigma_*/\sigma_n(A) < 1$.

1. Residual $\Rightarrow$ spectral separation. The Rayleigh quotient at $[x_{\mathrm{LS}}^\top, -1]^\top$ yields

$$\boxed{\delta \leq \frac{\|r_{\mathrm{LS}}\|_2}{\sigma_n(A)\sqrt{1 + \|x_{\mathrm{LS}}\|_2^2}} \leq \eta_r := \frac{\|r_{\mathrm{LS}}\|_2}{\sigma_n(A)}.}$$

Hence $\eta_r \leq 1 - c$ implies $\delta \leq 1 - c$: a sufficiently compatible LS system has the singular-value gap needed by TLS. A large residual gives no such guarantee.

2. Separation $\Rightarrow$ proximity. Subtract the LS and TLS normal equations:

$$(A^\top A - \sigma_*^2 I)(x_{\mathrm{TLS}} - x_{\mathrm{LS}}) = \sigma_*^2 x_{\mathrm{LS}}, \qquad \frac{\|x_{\mathrm{TLS}} - x_{\mathrm{LS}}\|_2}{\|x_{\mathrm{LS}}\|_2} \leq \frac{\delta^2}{1 - \delta^2}.$$

The paper also gives a data-aware bound with $\kappa_{\mathrm{eff}} = \|b\|_2/(\sigma_n(A)\|x_{\mathrm{LS}}\|_2)$ [Shan and Wei 2025]:

$$\frac{\|x_{\mathrm{TLS}} - x_{\mathrm{LS}}\|_2}{\|x_{\mathrm{LS}}\|_2} \leq \kappa_{\mathrm{eff}} \frac{\delta}{1 - \delta} \left(\sqrt{2} + \frac{\sigma_n(A)}{\|b\|_2}\right).$$

For a computed $\widehat{x}$, add its LS forward error $\|\widehat{x} - x_{\mathrm{LS}}\|_2$ before claiming proximity to TLS.

Algorithm: RQI-SPCGTLS

Input: $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$. **Output:** x_{TLS}. Here $\mathrm{SPCG}(A, y, \rho)$ solves $(A^\top A - \rho I)z = y$ using the QR factor of a sketch SA.

$x_0 \leftarrow \mathrm{FOSSIL}(A, b)$ — backward-stable LS seed
$r_0 \leftarrow b - Ax_0$
$\rho_0 \leftarrow \|r_0\|_2^2/(1 + \|x_0\|_2^2)$ — Rayleigh quotient
$u_0 \leftarrow \mathrm{SPCG}(A, x_0, 0)$ — first inverse step
$x_1 \leftarrow x_0 + \rho_0 u_0$ — inverse-iteration warm-up
for $k = 1, 2, \ldots$ **do**
  $r_k \leftarrow b - Ax_k$
  $\rho_k \leftarrow \|r_k\|_2^2/(1 + \|x_k\|_2^2)$ — update the shift
  $f_k \leftarrow -A^\top r_k - \rho_k x_k$
  $g_k \leftarrow -b^\top r_k + \rho_k$ — eigen-residual blocks
  $\omega_k \leftarrow \mathrm{SPCG}(A, -f_k, \rho_k)$ — shifted solve 1
  $z_k \leftarrow x_k + \omega_k$
  $\beta_k \leftarrow (z_k^\top f_k - g_k)/(1 + z_k^\top x_k)$ — scalar correction
  $u_k \leftarrow \mathrm{SPCG}(A, x_k, \rho_k)$ — shifted solve 2
  $x_{k+1} \leftarrow z_k + \beta_k u_k$ — RQI update
end for

The stable randomized LS initializer is described in [Epperly et al. 2026]; the RQI formulation builds on [Björck et al. 2000].

RQI-SPCGTLS: conditions for the inner solves

Let $SA = QR$ and use R as the SPCG preconditioner. The finite-precision result assumes [Shan and Wei 2025]:

1. S embeds $\mathrm{col}(A)$ with distortion $\eta \le 0.9$, and $\|\,\mathrm{fl}(SA) - SA\|_2 \lesssim \|A\|_2\, u$;
2. $\kappa_2(A)u \ll 1$ (numerical full rank);
3. $\delta := \sigma_{n+1}([A, b])/\sigma_n(A) \le 1 - c$ for a gap $c > 0$ independent of roundoff.

Why the ratio matters. At the limiting shift $\rho = \sigma_*^2$, the two preconditioned matrices are

$$G = R^{-\top} A^\top A R^{-1}, \qquad P = R^{-\top}(A^\top A - \rho I) R^{-1}.$$

The embedding gives $\frac{1}{1+\eta} \le \sigma_i(AR^{-1}) \le \frac{1}{1-\eta}$, so in exact arithmetic

$$\kappa_2(G) \le \left(\frac{1+\eta}{1-\eta}\right)^2, \qquad \kappa_2(P) \le \frac{(1+\eta)^2}{(1-\eta)^2(1-\delta^2)}.$$

Stable sketch formation and $\kappa_2(A)u \ll 1$ preserve these bounds up to small rounding terms. As $\delta \to 1$, the shifted system becomes hard even with a good sketch.

RQI-SPCGTLS: why the outer direction survives

Write $C = [A, b]$, $M = C^\top C$, and assume $\sigma_* > 0$. The finite-precision inverse step can be represented as

$$(M - H)y = q + f, \qquad My = q + g, \quad g = f + Hy,$$

where H and f collect rounding and inner-solve errors. The whitened-basis analysis follows [Epperly et al. 2026]. Under the stated conditions these errors are small; the resulting error is $e = M^{-1}g$.

Proof sketch. Let v_* be the right singular vector of C for σ_*, and decompose $g = \|g\|_2(v_* \cos\phi + w \sin\phi)$ with $w \perp v_*$. Because M^{-1} amplifies the target eigendirection most and $\sigma_n(C) \geq \sigma_n(A)$ by interlacing,

$$\tan \angle(e, v_*) \leq \left(\frac{\sigma_*}{\sigma_n(C)}\right)^2 \tan\phi \leq \delta^2 \tan\phi.$$

With a good LS seed, $|\tan\phi|$ is moderate. Thus even an amplified computational error points mainly toward v_* when the gap is clear; it does not redirect the inverse step. This is a direction-preservation argument, following the eigenvector perturbation viewpoint of [Parlett 1998]. The conclusion concerns the inverse phase. The RQI phase still requires an initial vector in its local basin and sufficiently accurate inner solves.

Stopping: eigen-residual and backward error

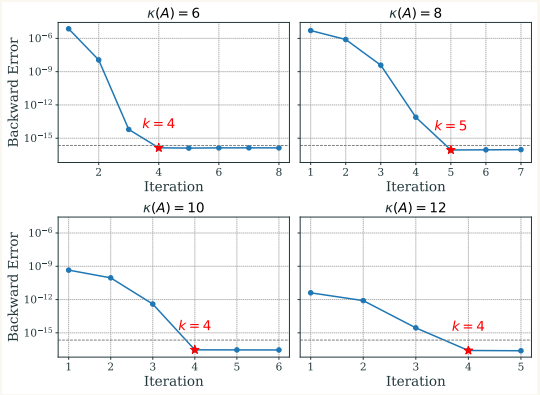
Published rule. Monitor the eigen-residual $\gamma_k = \|(f_k, g_k)\|_2 / \|q(x_k)\|_2$ and stop when it rises at the roundoff floor.

Companion certificate. Let $\tau_1(x_k)$ be the least data perturbation enforcing TLS stationarity. If S embeds $\mathrm{span}(C)$, then

$$\frac{\tau_1(x_k)}{\|C\|_{\mathrm{F}}} \le \underbrace{\frac{\sqrt{2}(1+\epsilon)\widetilde{\tau}_1^{\mathrm{sk}}(x_k)}{\|C\|_{\mathrm{F}}}}_{U_k}.$$

The test $U_k \le \mathrm{tol}$ certifies nearby stationarity. A retained singular-value gap identifies the TLS minimizer.

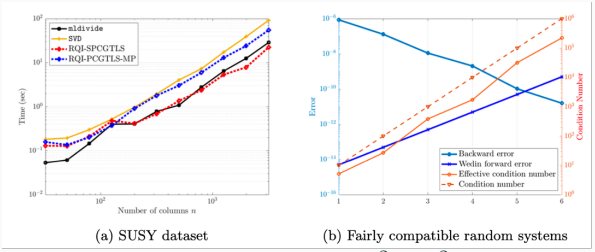
The illustrative curves reach their floor after about four or five outer steps.



Numerical evidence: SuiteSparse and SUSY

SuiteSparse BIBD matrices [Davis and Hu 2011]; solver time (s).

Matrix	m	n	nnz	SPCG	PCG-MP	Speedup
`bibd_12_5`	792	66	7,920	0.02	0.03	$1.5\times$
`bibd_17_8`	24,310	136	680,680	0.04	0.09	$2.3\times$
`bibd_20_10`	184,756	190	8,314,020	5.94	9.88	$1.6\times$
`bibd_22_8`	319,770	231	8,953,560	6.38	20.72	$3.3\times$



(a) SUSY dataset (b) Fairly compatible random systems

Figure. (a) SUSY kernel regression: $K(z, z') = \exp(-\|z - z'\|_2^2/(2\sigma^2))$, $\sigma = 4$; (b) fairly compatible 500×20 systems with varying conditioning.

Observed: up to $3.3\times$ faster on BIBD; small backward error persists as conditioning grows.

Conclusion

1. Backward analysis $\rightarrow$ a computable measure. The optimal mTLS backward error has an explicit expression and tight sketched estimates. These estimates provide stopping certificates and may also guide null-space estimation.

2. LS–TLS connection $\rightarrow$ an iterative solver. For sufficiently compatible data, a forward-accurate LS solution is a useful TLS seed. RQI-SPCGTLS uses the resulting spectral gap to make randomized preconditioning effective. Open question: Can we develop a backward-stable TLS solver for ill-conditioned, large-scale systems, with guarantees that extend to null-space estimation?

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